Mock study: oral microbiome shifts in experimental gingivitis

Abstract
We profiled supragingival plaque microbiomes of 40 subjects across a longitudinal gingivitis intervention to identify disease-associated taxa.

Methods
Microbiome profiling. Genus-level abundance tables were obtained by 16S rRNA gene sequencing. Community structure was examined by principal coordinates analysis (PCoA) on Bray-Curtis dissimilarities, with samples colored by disease group. Group effects on overall composition were tested by PERMANOVA (adonis, 999 permutations, stratified by subject). Differentially abundant genera between baseline and later visits were identified after centered log-ratio (CLR) transformation using the Wilcoxon rank-sum test with Benjamini-Hochberg correction. Genus-genus co-occurrence networks were inferred with Spearman correlations. Functional potential was predicted from 16S profiles with PICRUSt2 against the KEGG database. Figures and a summary report were compiled for all analyses.